FLATNESS AFTER CUTTING A FLAT ALGEBRA BY A QUASI-REGULAR SEQUENCE

THE STACKS PROJECT AUTHORS

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Allowed standard prerequisites (editorial).

- Flatness is local on the source and the target. Algebra, Lemma \ref{algebra-lemma-flat-localization}, cited at proof lines 8041-8042. Reduces the claim to flatness of a localised ring map.
- The Grothendieck criterion: a quotient by a sequence is flat when the sequence maps to a regular sequence in the fibre. Algebra, Lemma \ref{algebra-lemma-grothendieck-regular-sequence-general}, cited at proof lines 8043-8045. Converts the required flatness into a regularity statement in the fibre.
- Quasi-regularity is preserved when passing to the fibre over the residue field. Lemma \ref{lemma-relative-regular-immersion-algebra}, cited at proof lines 8051-8054. Produces the quasi-regular sequence in the fibre of the quotient.
- A quasi-regular sequence in a Noetherian local ring is a regular sequence, and finite modules cut out by a regular sequence are finite free. Lemma \ref{lemma-noetherian-finite-all-equivalent}, cited at proof lines 8055-8057. Upgrades quasi-regularity in the fibre to regularity, which is what the Grothendieck criterion needs.

Source reference. Original lemma 0CEQ: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 15585–15585. The lemma's own canonical tag, quoted so that readers can resolve the label inside the delivered text; it points at the counted statement itself and carries no proof content.

Source reference. Original lemma 00HT: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 655–655. Navigation pointer to a statement the

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counted proof refers to but does not contain; nothing load-bearing is delegated to it.

Source reference. Original lemma 0470: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 5491–5491. Navigation pointer to a statement the counted proof refers to but does not contain; nothing load-bearing is delegated to it.

Source reference. Original lemma 0CEP: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 15584–15584. Navigation pointer to a statement the counted proof refers to but does not contain; nothing load-bearing is delegated to it.

Source reference. Original lemma 09CC: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 11825–11825. Navigation pointer to a statement the counted proof refers to but does not contain; nothing load-bearing is delegated to it.

Lemma 0.2. *Let $A' \to B'$ be a ring map. Let $I \subset A'$ be an ideal. Set $A = A'/I$ and $B = B'/IB'$. Let $f'_1, \ldots, f'_r \in B'$. Assume*

1. *$A' \to B'$ is flat and of finite presentation,*
2. *I is locally nilpotent,*
3. *the images $f_1, \ldots, f_r \in B$ form a quasi-regular sequence,*
4. *$B/(f_1, \ldots, f_r)$ is flat over A.*

Then $B'/(f'_1, \ldots, f'_r)$ is flat over A'.

Proof. Set $C' = B'/(f'_1, \ldots, f'_r)$. We have to show $A' \to C'$ is flat. Let $\mathfrak{r}' \subset C'$ be a prime ideal lying over $\mathfrak{p}' \subset A'$. We let $\mathfrak{q}' \subset B'$ be the inverse image of $\mathfrak{r}'$. By Algebra, Lemma 00HT it suffices to show that $A'_{\mathfrak{p}'} \to C'_{\mathfrak{q}'}$ is flat. Algebra, Lemma 0470 tells us it suffices to show that $f'_1, \ldots, f'_r$ map to a regular sequence in

$$B'_{\mathfrak{q}'}/\mathfrak{p}'B'_{\mathfrak{q}'} = B_{\mathfrak{q}}/\mathfrak{p}B_{\mathfrak{q}} = (B \otimes_A \kappa(\mathfrak{p}))_{\mathfrak{q}}$$

with obvious notation. What we know is that $f_1, \ldots, f_r$ is a quasi-regular sequence in B and that $B/(f_1, \ldots, f_r)$ is flat over A. By Lemma 0CEP the images $\overline{f}_1, \ldots, \overline{f}_r$ of $f'_1, \ldots, f'_r$ in $B \otimes_A \kappa(\mathfrak{p})$ form a quasi-regular sequence. Since $(B \otimes_A \kappa(\mathfrak{p}))_{\mathfrak{q}}$ is a Noetherian local ring, we conclude by Lemma 09CC. $\square$